Drinfeld level structures via prismatic Dieudonne theory

Gyujin Oh

Oct 19, 2022

Outline

1. Integral models of Shimura varieties
2. (Prismatic) Dieudonné theory
3. Level structures via (prismatic) Dieudonné theory

Modular curves

Modular curves first arise as certain arithmetic quotients of the upper half plane $\mathbb{H}$. More precisely, $\mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ via Möbius transformation $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) z = \frac{az+b}{cz+d}$, and modular curves are $\Gamma\backslash\mathbb{H}$, where $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ is a **congruence subgroup** such as

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}$$

Modular curves

Modular curves first arise as certain arithmetic quotients of the upper half plane $\mathbb{H}$. More precisely, $\mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ via Möbius transformation $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) z = \frac{az+b}{cz+d}$, and modular curves are $\Gamma\backslash\mathbb{H}$, where $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ is a **congruence subgroup** such as

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}$$

$$\subset \quad \Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}$$

Modular curves

Modular curves first arise as certain arithmetic quotients of the upper half plane $\mathbb{H}$. More precisely, $\mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ via Möbius transformation $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) z = \frac{az+b}{cz+d}$, and modular curves are $\Gamma\backslash\mathbb{H}$, where $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ is a **congruence subgroup** such as

$$\begin{aligned} & \Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\} \\ \subset \quad & \Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod{N} \right\} \\ \subset \quad & \Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{N} \right\} \end{aligned}$$

Modular curves

Modular curves first arise as certain arithmetic quotients of the upper half plane $\mathbb{H}$. More precisely, $\mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ via Möbius transformation $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) z = \frac{az+b}{cz+d}$, and modular curves are $\Gamma\backslash\mathbb{H}$, where $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ is a **congruence subgroup** such as

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}$$

$$\subset \quad \Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}$$

$$\subset \quad \Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{N} \right\}$$

Such quotient can be regarded as the moduli space of elliptic curves equipped with a certain structure called a **level structure**.

Modular curves

Modular curves first arise as certain arithmetic quotients of the upper half plane $\mathbb{H}$. More precisely, $\mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ via Möbius transformation $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) z = \frac{az+b}{cz+d}$, and modular curves are $\Gamma\backslash\mathbb{H}$, where $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ is a **congruence subgroup** such as

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}$$

$$\subset \quad \Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}$$

$$\subset \quad \Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{N} \right\}$$

Such quotient can be regarded as the moduli space of elliptic curves equipped with a certain structure called a **level structure**.In particular, if one could make sense of the notion of level structures over more general base ring R (other than $\mathbb{C}$), we will obtain a "canonical model" of the modular curve over R.

Level structures on elliptic curves

Over $\mathbb{C}$, the N-torsion points $E[N]$ of an elliptic curve E form a group that is isomorphic to $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. A structure that rigidifies this isomorphism to a certain extent is in general called a **level structure**.

Level structures on elliptic curves

Over $\mathbb{C}$, the N-torsion points $E[N]$ of an elliptic curve E form a group that is isomorphic to $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. A structure that rigidifies this isomorphism to a certain extent is in general called a **level structure**.

Example

$\Gamma(N)$ corresponds to fixing the isomorphism $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$.

Level structures on elliptic curves

Over $\mathbb{C}$, the N-torsion points $E[N]$ of an elliptic curve E form a group that is isomorphic to $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. A structure that rigidifies this isomorphism to a certain extent is in general called a **level structure**.

Example

$\Gamma(N)$ corresponds to fixing the isomorphism $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$.
$\Gamma_0(N)$ corresponds to fixing a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.

Level structures on elliptic curves

Over $\mathbb{C}$, the N-torsion points $E[N]$ of an elliptic curve E form a group that is isomorphic to $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. A structure that rigidifies this isomorphism to a certain extent is in general called a **level structure**.

Example

$\Gamma(N)$ corresponds to fixing the isomorphism $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$.
$\Gamma_0(N)$ corresponds to fixing a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.
$\Gamma_1(N)$ corresponds to fixing a generator of a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.

Level structures on elliptic curves

Over $\mathbb{C}$, the N-torsion points $E[N]$ of an elliptic curve E form a group that is isomorphic to $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. A structure that rigidifies this isomorphism to a certain extent is in general called a **level structure**.

Example

$\Gamma(N)$ corresponds to fixing the isomorphism $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$.
$\Gamma_0(N)$ corresponds to fixing a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.
$\Gamma_1(N)$ corresponds to fixing a generator of a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.

These kinds of level structures work for an elliptic curve over a general base S, as long as the structure of $E[N]$ is $(\mathbb{Z}/N\mathbb{Z})^2$. This is the case if N is invertible on S; namely, if S is a $\mathbb{Z}[1/N]$-scheme.

Level structures on elliptic curves

Over $\mathbb{C}$, the N-torsion points $E[N]$ of an elliptic curve E form a group that is isomorphic to $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. A structure that rigidifies this isomorphism to a certain extent is in general called a **level structure**.

Example

$\Gamma(N)$ corresponds to fixing the isomorphism $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$.
$\Gamma_0(N)$ corresponds to fixing a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.
$\Gamma_1(N)$ corresponds to fixing a generator of a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.

These kinds of level structures work for an elliptic curve over a general base S, as long as the structure of $E[N]$ is $(\mathbb{Z}/N\mathbb{Z})^2$. This is the case if N is invertible on S; namely, if S is a $\mathbb{Z}[1/N]$-scheme. On the other hand, if N is not invertible on S, the structure of $E[N]$ could be more complicated.

Level structures on elliptic curves

Over $\mathbb{C}$, the N-torsion points $E[N]$ of an elliptic curve E form a group that is isomorphic to $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. A structure that rigidifies this isomorphism to a certain extent is in general called a **level structure**.

Example

$\Gamma(N)$ corresponds to fixing the isomorphism $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$.
$\Gamma_0(N)$ corresponds to fixing a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.
$\Gamma_1(N)$ corresponds to fixing a generator of a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.

These kinds of level structures work for an elliptic curve over a general base S, as long as the structure of $E[N]$ is $(\mathbb{Z}/N\mathbb{Z})^2$. This is the case if N is invertible on S; namely, if S is a $\mathbb{Z}[1/N]$-scheme. On the other hand, if N is not invertible on S, the structure of $E[N]$ could be more complicated.

Example

Let $S = \operatorname{Spec} \overline{\mathbb{F}}_p$. Then, the structure of $E[p]$ depends on whether E is **ordinary** or **supersingular**.

Level structures on elliptic curves

Over $\mathbb{C}$, the N-torsion points $E[N]$ of an elliptic curve E form a group that is isomorphic to $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. A structure that rigidifies this isomorphism to a certain extent is in general called a **level structure**.

Example

$\Gamma(N)$ corresponds to fixing the isomorphism $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$.
$\Gamma_0(N)$ corresponds to fixing a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.
$\Gamma_1(N)$ corresponds to fixing a generator of a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.

These kinds of level structures work for an elliptic curve over a general base S, as long as the structure of $E[N]$ is $(\mathbb{Z}/N\mathbb{Z})^2$. This is the case if N is invertible on S; namely, if S is a $\mathbb{Z}[1/N]$-scheme. On the other hand, if N is not invertible on S, the structure of $E[N]$ could be more complicated.

Example

Let $S = \operatorname{Spec} \overline{\mathbb{F}}_p$. Then, the structure of $E[p]$ depends on whether E is **ordinary** or **supersingular**.If E is ordinary, $E[p] \cong \mathbb{Z}/p\mathbb{Z} \times \mu_p$, where μ_p is the multiplicative group scheme $\operatorname{Spec} \overline{\mathbb{F}}_p[t]/(t^p - 1)$.

Level structures on elliptic curves

Over $\mathbb{C}$, the N-torsion points $E[N]$ of an elliptic curve E form a group that is isomorphic to $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. A structure that rigidifies this isomorphism to a certain extent is in general called a **level structure**.

Example

$\Gamma(N)$ corresponds to fixing the isomorphism $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$.
$\Gamma_0(N)$ corresponds to fixing a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.
$\Gamma_1(N)$ corresponds to fixing a generator of a subgroup $(\mathbb{Z}/N\mathbb{Z}) \subset E[N]$.

These kinds of level structures work for an elliptic curve over a general base S, as long as the structure of $E[N]$ is $(\mathbb{Z}/N\mathbb{Z})^2$. This is the case if N is invertible on S; namely, if S is a $\mathbb{Z}[1/N]$-scheme. On the other hand, if N is not invertible on S, the structure of $E[N]$ could be more complicated.

Example

Let $S = \operatorname{Spec} \overline{\mathbb{F}}_p$. Then, the structure of $E[p]$ depends on whether E is **ordinary** or **supersingular**.If E is ordinary, $E[p] \cong \mathbb{Z}/p\mathbb{Z} \times \mu_p$, where μ_p is the multiplicative group scheme $\operatorname{Spec} \overline{\mathbb{F}}_p[t]/(t^p - 1)$.If E is supersingular, $E[p]$ is an extension of α_p by α_p, where α_p is the additive group scheme $\operatorname{Spec} \overline{\mathbb{F}}_p[t]/(t^p)$.

Drinfeld–Katz–Mazur level structures

It is quite unclear how to extend the usual definition to work over a general base. Inspired by Drinfeld, Katz and Mazur defined a working definition of level structures over a general base S.

Drinfeld–Katz–Mazur level structures

It is quite unclear how to extend the usual definition to work over a general base. Inspired by Drinfeld, Katz and Mazur defined a working definition of level structures over a general base S.

Definition

For an elliptic curve E over S and a finite abelian group A, an A-**structure of** E/S is a group homomorphism $\phi : A \to E(S)$ such that the effective Cartier divisor $\sum_{a \in A}[\phi(a)]$ defines a subgroup scheme of E/S.

Drinfeld–Katz–Mazur level structures

It is quite unclear how to extend the usual definition to work over a general base. Inspired by Drinfeld, Katz and Mazur defined a working definition of level structures over a general base S.

Definition

For an elliptic curve E over S and a finite abelian group A, an A-**structure of** E/S is a group homomorphism $\phi : A \to E(S)$ such that the effective Cartier divisor $\sum_{a \in A}[\phi(a)]$ defines a subgroup scheme of E/S.

Example

For $A = \mathbb{Z}/p\mathbb{Z}$ and E a supersingular elliptic curve over $\overline{\mathbb{F}}_p$, since $E[p](\overline{\mathbb{F}}_p)$ is trivial, there is only one $\mathbb{Z}/p\mathbb{Z}$-structure of $E/\overline{\mathbb{F}}_p$, namely $\phi : \mathbb{Z}/p\mathbb{Z} \to E[p](\overline{\mathbb{F}}_p)$ that sends every number to the identity.

Drinfeld–Katz–Mazur level structures

It is quite unclear how to extend the usual definition to work over a general base. Inspired by Drinfeld, Katz and Mazur defined a working definition of level structures over a general base S.

Definition

For an elliptic curve E over S and a finite abelian group A, an A-**structure of** E/S is a group homomorphism $\phi : A \to E(S)$ such that the effective Cartier divisor $\sum_{a \in A}[\phi(a)]$ defines a subgroup scheme of E/S.

Example

For $A = \mathbb{Z}/p\mathbb{Z}$ and E a supersingular elliptic curve over $\overline{\mathbb{F}}_p$, since $E[p](\overline{\mathbb{F}}_p)$ is trivial, there is only one $\mathbb{Z}/p\mathbb{Z}$-structure of $E/\overline{\mathbb{F}}_p$, namely $\phi : \mathbb{Z}/p\mathbb{Z} \to E[p](\overline{\mathbb{F}}_p)$ that sends every number to the identity. On the other hand, there are $(p-1)$ different $\mathbb{Z}/p\mathbb{Z}$ structures for $E/(\overline{\mathbb{F}}_p[\varepsilon]/\varepsilon^2)$.

Drinfeld–Katz–Mazur level structures

It is quite unclear how to extend the usual definition to work over a general base. Inspired by Drinfeld, Katz and Mazur defined a working definition of level structures over a general base S.

Definition

For an elliptic curve E over S and a finite abelian group A, an A-**structure of** E/S is a group homomorphism $\phi : A \to E(S)$ such that the effective Cartier divisor $\sum_{a \in A}[\phi(a)]$ defines a subgroup scheme of E/S.

Example

For $A = \mathbb{Z}/p\mathbb{Z}$ and E a supersingular elliptic curve over $\overline{\mathbb{F}}_p$, since $E[p](\overline{\mathbb{F}}_p)$ is trivial, there is only one $\mathbb{Z}/p\mathbb{Z}$-structure of $E/\overline{\mathbb{F}}_p$, namely $\phi : \mathbb{Z}/p\mathbb{Z} \to E[p](\overline{\mathbb{F}}_p)$ that sends every number to the identity. On the other hand, there are $(p-1)$ different $\mathbb{Z}/p\mathbb{Z}$ structures for $E/(\overline{\mathbb{F}}_p[\varepsilon]/\varepsilon^2)$.

This turns out to be the correct notion of level structures for elliptic curves over arbitrary base ring, and this defines the canonical integral models of modular curves of the aforementioned levels over $\mathbb{Z}$ that are regular.

Drinfeld–Katz–Mazur level structures, cont'd

The definition of Katz and Mazur relied crucially on the fact that E is one-dimensional.

Drinfeld–Katz–Mazur level structures, cont'd

The definition of Katz and Mazur relied crucially on the fact that E is one-dimensional. There is an intrinsic characterization of Drinfeld level structures for a finite flat group scheme $E[N]$ without referring to its embedding into E, but this level structure turns out to be very bad in general.

Drinfeld–Katz–Mazur level structures, cont'd

The definition of Katz and Mazur relied crucially on the fact that E is one-dimensional. There is an intrinsic characterization of Drinfeld level structures for a finite flat group scheme $E[N]$ without referring to its embedding into E, but this level structure turns out to be very bad in general. Even for the next simplest example of $(\mathbb{Z}/p\mathbb{Z})^2$-structures on $\mu_p \times \mu_p$ over $\mathbb{Z}_p$, the moduli scheme of such level structures is not even $\mathbb{Z}_p$-flat (i.e. has a p-torsion).

Drinfeld–Katz–Mazur level structures, cont'd

The definition of Katz and Mazur relied crucially on the fact that E is one-dimensional. There is an intrinsic characterization of Drinfeld level structures for a finite flat group scheme $E[N]$ without referring to its embedding into E, but this level structure turns out to be very bad in general. Even for the next simplest example of $(\mathbb{Z}/p\mathbb{Z})^2$-structures on $\mu_p \times \mu_p$ over $\mathbb{Z}_p$, the moduli scheme of such level structures is not even $\mathbb{Z}_p$-flat (i.e. has a p-torsion).

The moduli of elliptic curves $Y(1)$ over $\mathbb{C}$ has a natural integral model $\mathfrak{Y}(1)$ over $\mathbb{Z}$ as the moduli of elliptic curves, and the modular curve $Y(N)$ over $\mathbb{C}$ has a "canonical" integral model $\mathfrak{Y}(N)$ over $\mathbb{Z}$ by taking

$$\mathfrak{Y}(N) := \mathfrak{E}[N]^{\text{DKM-}(\mathbb{Z}/N\mathbb{Z})^2\text{-structure}}$$

where $\mathfrak{E}/\mathfrak{Y}(1)$ is the universal elliptic curve.

Drinfeld–Katz–Mazur level structures, cont'd

The definition of Katz and Mazur relied crucially on the fact that E is one-dimensional. There is an intrinsic characterization of Drinfeld level structures for a finite flat group scheme $E[N]$ without referring to its embedding into E, but this level structure turns out to be very bad in general. Even for the next simplest example of $(\mathbb{Z}/p\mathbb{Z})^2$-structures on $\mu_p \times \mu_p$ over $\mathbb{Z}_p$, the moduli scheme of such level structures is not even $\mathbb{Z}_p$-flat (i.e. has a p-torsion).

The moduli of elliptic curves $Y(1)$ over $\mathbb{C}$ has a natural integral model $\mathfrak{Y}(1)$ over $\mathbb{Z}$ as the moduli of elliptic curves, and the modular curve $Y(N)$ over $\mathbb{C}$ has a "canonical" integral model $\mathfrak{Y}(N)$ over $\mathbb{Z}$ by taking

$$\mathfrak{Y}(N) := \mathfrak{E}[N]^{\text{DKM-}(\mathbb{Z}/N\mathbb{Z})^2\text{-structure}}$$

where $\mathfrak{E}/\mathfrak{Y}(1)$ is the universal elliptic curve.

The deficiency of Drinfeld–Katz–Mazur level structure alluded above implies that the integral model $\mathfrak{A}_g(N)$ over $\mathbb{Z}$ obtained from the universal abelian variety over the moduli of principally polarized abelian varieties $\mathfrak{A}_g(1)$ over $\mathbb{Z}$ is not even flat over $\mathfrak{A}_g(1)$.

Integral canonical models of Shimura varieties

The modular curves $Y(N)$, and more generally $A_g(N)$, are examples of **Shimura varieties**. Those are central objects in the Langlands program, and they have canonical models in a very precise sense. We expect that Shimura varieties are moduli spaces of motives with additional structures, there must be "integral canonical models".

Integral canonical models of Shimura varieties

The modular curves $Y(N)$, and more generally $A_g(N)$, are examples of **Shimura varieties**. Those are central objects in the Langlands program, and they have canonical models in a very precise sense. We expect that Shimura varieties are moduli spaces of motives with additional structures, there must be "integral canonical models".

- Modular curves: $G = \mathrm{SL}_2(\mathbb{Q})$
  Elliptic curves over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$

Integral canonical models of Shimura varieties

The modular curves $Y(N)$, and more generally $A_g(N)$, are examples of **Shimura varieties**. Those are central objects in the Langlands program, and they have canonical models in a very precise sense. We expect that Shimura varieties are moduli spaces of motives with additional structures, there must be "integral canonical models".

- Modular curves: $G = \mathrm{SL}_2(\mathbb{Q})$
  Elliptic curves over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Siegel modular varieties: $G = \mathrm{Sp}_{2n}(\mathbb{Q})$
  Principally polarized abelian varieties over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$

Integral canonical models of Shimura varieties

The modular curves $Y(N)$, and more generally $A_g(N)$, are examples of **Shimura varieties**. Those are central objects in the Langlands program, and they have canonical models in a very precise sense. We expect that Shimura varieties are moduli spaces of motives with additional structures, there must be "integral canonical models".

- Modular curves: $G = \mathrm{SL}_2(\mathbb{Q})$
  Elliptic curves over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Siegel modular varieties: $G = \mathrm{Sp}_{2n}(\mathbb{Q})$
  Principally polarized abelian varieties over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Hilbert modular varieties: $G = \mathrm{SL}_2(K)$, K totally real of degree d
  Abelian varieties with real multiplication by $\mathcal{O}_K$ over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$

Integral canonical models of Shimura varieties

The modular curves $Y(N)$, and more generally $A_g(N)$, are examples of **Shimura varieties**. Those are central objects in the Langlands program, and they have canonical models in a very precise sense. We expect that Shimura varieties are moduli spaces of motives with additional structures, there must be "integral canonical models".

- Modular curves: $G = \mathrm{SL}_2(\mathbb{Q})$
  Elliptic curves over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Siegel modular varieties: $G = \mathrm{Sp}_{2n}(\mathbb{Q})$
  Principally polarized abelian varieties over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Hilbert modular varieties: $G = \mathrm{SL}_2(K)$, K totally real of degree d
  Abelian varieties with real multiplication by $\mathcal{O}_K$ over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Picard modular surfaces: $G = \mathrm{U}(2,1)$, K imaginary quadratidc
  Abelian threefolds with complex multiplication by $\mathcal{O}_K$ with "signature $(2,1)$" over $K \rightsquigarrow \mathcal{O}_K$

Integral canonical models of Shimura varieties

The modular curves $Y(N)$, and more generally $A_g(N)$, are examples of **Shimura varieties**. Those are central objects in the Langlands program, and they have canonical models in a very precise sense. We expect that Shimura varieties are moduli spaces of motives with additional structures, there must be "integral canonical models".

- Modular curves: $G = \mathrm{SL}_2(\mathbb{Q})$
  Elliptic curves over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Siegel modular varieties: $G = \mathrm{Sp}_{2n}(\mathbb{Q})$
  Principally polarized abelian varieties over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Hilbert modular varieties: $G = \mathrm{SL}_2(K)$, K totally real of degree d
  Abelian varieties with real multiplication by $\mathcal{O}_K$ over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Picard modular surfaces: $G = \mathrm{U}(2,1)$, K imaginary quadratidc
  Abelian threefolds with complex multiplication by $\mathcal{O}_K$ with "signature $(2,1)$" over $K \rightsquigarrow \mathcal{O}_K$

The progress on integral canonical models has been somewhat limited, because

- not all Shimura varieties have moduli interpretations,

Integral canonical models of Shimura varieties

The modular curves $Y(N)$, and more generally $A_g(N)$, are examples of **Shimura varieties**. Those are central objects in the Langlands program, and they have canonical models in a very precise sense. We expect that Shimura varieties are moduli spaces of motives with additional structures, there must be "integral canonical models".

- Modular curves: $G = \mathrm{SL}_2(\mathbb{Q})$
  Elliptic curves over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Siegel modular varieties: $G = \mathrm{Sp}_{2n}(\mathbb{Q})$
  Principally polarized abelian varieties over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Hilbert modular varieties: $G = \mathrm{SL}_2(K)$, K totally real of degree d
  Abelian varieties with real multiplication by $\mathcal{O}_K$ over $\mathbb{Q} \rightsquigarrow \mathbb{Z}$
- Picard modular surfaces: $G = \mathrm{U}(2,1)$, K imaginary quadratidc
  Abelian threefolds with complex multiplication by $\mathcal{O}_K$ with "signature $(2,1)$" over $K \rightsquigarrow \mathcal{O}_K$

The progress on integral canonical models has been somewhat limited, because

- not all Shimura varieties have moduli interpretations,
- and the level structures do not properly generalize.

Integral canonical models of Shimura varieties, cont'd

Let us focus on the second problem, so let us restrict to the Shimura varieties that have interpretations as moduli of certain abelian varieties.

Integral canonical models of Shimura varieties, cont'd

Let us focus on the second problem, so let us restrict to the Shimura varieties that have interpretations as moduli of certain abelian varieties.

Some known cases (Harris–Taylor, de Jong, Lan, Rapoport–Zink, Kisin, Vasiu, Pappas, Haines–Rapoport, $\cdots$):

Integral canonical models of Shimura varieties, cont'd

Let us focus on the second problem, so let us restrict to the Shimura varieties that have interpretations as moduli of certain abelian varieties.

Some known cases (Harris–Taylor, de Jong, Lan, Rapoport–Zink, Kisin, Vasiu, Pappas, Haines–Rapoport, $\cdots$):

- The Drinfeld–Katz–Mazur level structures work equally well for 1-dimensional p-divisible groups. In particular, for abelian varieties with complex multiplication with signature $(n, 1)$, there is a part of the p-divisible group that is 1-dimensional.

Integral canonical models of Shimura varieties, cont'd

Let us focus on the second problem, so let us restrict to the Shimura varieties that have interpretations as moduli of certain abelian varieties.

Some known cases (Harris–Taylor, de Jong, Lan, Rapoport–Zink, Kisin, Vasiu, Pappas, Haines–Rapoport, $\cdots$):

- The Drinfeld–Katz–Mazur level structures work equally well for 1-dimensional p-divisible groups. In particular, for abelian varieties with complex multiplication with signature $(n, 1)$, there is a part of the p-divisible group that is 1-dimensional. Canonical integral models of the corresponding Shimura varieties ($G = \mathrm{U}(n, 1)$) are constructed for arbitrarily deep level.

Integral canonical models of Shimura varieties, cont'd

Let us focus on the second problem, so let us restrict to the Shimura varieties that have interpretations as moduli of certain abelian varieties.

Some known cases (Harris–Taylor, de Jong, Lan, Rapoport–Zink, Kisin, Vasiu, Pappas, Haines–Rapoport, $\cdots$):

- The Drinfeld–Katz–Mazur level structures work equally well for 1-dimensional p-divisible groups. In particular, for abelian varieties with complex multiplication with signature $(n, 1)$, there is a part of the p-divisible group that is 1-dimensional. Canonical integral models of the corresponding Shimura varieties ($G = \mathrm{U}(n, 1)$) are constructed for arbitrarily deep level.
- For $A_{g,\Gamma_{\mathbf{o}}(p)}$, there is a natural integral model given as the moduli space of chain of isogenies,

Integral canonical models of Shimura varieties, cont'd

Let us focus on the second problem, so let us restrict to the Shimura varieties that have interpretations as moduli of certain abelian varieties.

Some known cases (Harris–Taylor, de Jong, Lan, Rapoport–Zink, Kisin, Vasiu, Pappas, Haines–Rapoport, $\cdots$):

- The Drinfeld–Katz–Mazur level structures work equally well for 1-dimensional p-divisible groups. In particular, for abelian varieties with complex multiplication with signature $(n, 1)$, there is a part of the p-divisible group that is 1-dimensional. Canonical integral models of the corresponding Shimura varieties ($G = \mathrm{U}(n, 1)$) are constructed for arbitrarily deep level.
- For $A_{g,\Gamma_0(p)}$, there is a natural integral model given as the moduli space of chain of isogenies,

$$S \mapsto \{(A_0 \xrightarrow{\alpha} \cdots \xrightarrow{\alpha} A_g, \lambda_0, \lambda_g)\},$$

Integral canonical models of Shimura varieties, cont'd

Let us focus on the second problem, so let us restrict to the Shimura varieties that have interpretations as moduli of certain abelian varieties.

Some known cases (Harris–Taylor, de Jong, Lan, Rapoport–Zink, Kisin, Vasiu, Pappas, Haines–Rapoport, $\cdots$):

- The Drinfeld–Katz–Mazur level structures work equally well for 1-dimensional p-divisible groups. In particular, for abelian varieties with complex multiplication with signature $(n, 1)$, there is a part of the p-divisible group that is 1-dimensional. Canonical integral models of the corresponding Shimura varieties ($G = \mathrm{U}(n, 1)$) are constructed for arbitrarily deep level.
- For $A_{g,\Gamma_0(p)}$, there is a natural integral model given as the moduli space of chain of isogenies,

$$S \mapsto \{(A_0 \xrightarrow{\alpha} \cdots \xrightarrow{\alpha} A_g, \lambda_0, \lambda_g)\},$$

  where $A_i \xrightarrow{\alpha} A_{i+1}$ is an isogeny of degree p, and λ_0, λ_g are principal polarizations of A_0, A_g that satisfy certain conditions with respect to Weil pairing.

Integral canonical models of Shimura varieties, cont'd

- In general, canonical integral models of Shimura varieties with **parahoric level** can all be thought as coming from the above integral model of $A_{g,\Gamma_{\mathbf{o}}(p)}$.

Integral canonical models of Shimura varieties, cont'd

- In general, canonical integral models of Shimura varieties with **parahoric level** can all be thought as coming from the above integral model of $A_{g,\Gamma_{\mathbf{o}}(p)}$. Here, a **parahoric subgroup** for a quasisplit reductive group $\mathcal{G}/\mathbb{Z}_p$ is the preimage of a parabolic subgroup of $\mathcal{G}(\mathbb{F}_p)$ via the quotient map $\mathcal{G}(\mathbb{Z}_p) \twoheadrightarrow \mathcal{G}(\mathbb{F}_p)$.

Integral canonical models of Shimura varieties, cont'd

- In general, canonical integral models of Shimura varieties with **parahoric level** can all be thought as coming from the above integral model of $A_{g,\Gamma_{\mathbf{o}}(p)}$. Here, a **parahoric subgroup** for a quasisplit reductive group $\mathcal{G}/\mathbb{Z}_p$ is the preimage of a parabolic subgroup of $\mathcal{G}(\mathbb{F}_p)$ via the quotient map $\mathcal{G}(\mathbb{Z}_p) \twoheadrightarrow \mathcal{G}(\mathbb{F}_p)$.
- There have been very few works towards integral models of Shimura varieties other than parahoric levels. One successful account is the integral model for $\Gamma_1(p)$ level, using $\Gamma_0(p)$ level + the Oort–Tate theory of classification of finite flat group schemes of order p.

Integral canonical models of Shimura varieties, cont'd

- In general, canonical integral models of Shimura varieties with **parahoric level** can all be thought as coming from the above integral model of $A_{g,\Gamma_{\mathbf{o}}(p)}$. Here, a **parahoric subgroup** for a quasisplit reductive group $\mathcal{G}/\mathbb{Z}_p$ is the preimage of a parabolic subgroup of $\mathcal{G}(\mathbb{F}_p)$ via the quotient map $\mathcal{G}(\mathbb{Z}_p) \twoheadrightarrow \mathcal{G}(\mathbb{F}_p)$.
- There have been very few works towards integral models of Shimura varieties other than parahoric levels. One successful account is the integral model for $\Gamma_1(p)$ level, using $\Gamma_0(p)$ level + the Oort–Tate theory of classification of finite flat group schemes of order p.
  In particular, over a $\mathbb{Z}_p$-algebra R, a finite flat group scheme G of rank p is, as a scheme, of the form $\operatorname{Spec} R[x]/(x^p - ax)$ for some $a \in R$, and one can define the scheme of **Oort–Tate generators** as $\operatorname{Spec} R[x]/(x^{p-1} - a)$.

Integral canonical models of Shimura varieties, cont'd

- In general, canonical integral models of Shimura varieties with **parahoric level** can all be thought as coming from the above integral model of $A_{g,\Gamma_{\mathbf{o}}(p)}$. Here, a **parahoric subgroup** for a quasisplit reductive group $\mathcal{G}/\mathbb{Z}_p$ is the preimage of a parabolic subgroup of $\mathcal{G}(\mathbb{F}_p)$ via the quotient map $\mathcal{G}(\mathbb{Z}_p) \twoheadrightarrow \mathcal{G}(\mathbb{F}_p)$.
- There have been very few works towards integral models of Shimura varieties other than parahoric levels. One successful account is the integral model for $\Gamma_1(p)$ level, using $\Gamma_0(p)$ level + the Oort–Tate theory of classification of finite flat group schemes of order p.
  In particular, over a $\mathbb{Z}_p$-algebra R, a finite flat group scheme G of rank p is, as a scheme, of the form $\operatorname{Spec} R[x]/(x^p - ax)$ for some $a \in R$, and one can define the scheme of **Oort–Tate generators** as $\operatorname{Spec} R[x]/(x^{p-1} - a)$.

In this work, we try to give a more streamlined approach towards all level structures using (prismatic) Dieudonné theory.

Outline

1. Integral models of Shimura varieties
2. (Prismatic) Dieudonné theory
3. Level structures via (prismatic) Dieudonné theory

Dieudonné theory

Dieudonné theory gives a classification of p-divisible groups and finite flat group schemes over a perfect field k of characteristic p in terms of semi-linear algebraic data.

Dieudonné theory

Dieudonné theory gives a classification of p-divisible groups and finite flat group schemes over a perfect field k of characteristic p in terms of semi-linear algebraic data.

Classical Dieudonné theory (Dieudonné, Grothendieck, Messing, Mazur)

There is an anti-equivalence of categories

Dieudonné theory

Dieudonné theory gives a classification of p-divisible groups and finite flat group schemes over a perfect field k of characteristic p in terms of semi-linear algebraic data.

Classical Dieudonné theory (Dieudonné, Grothendieck, Messing, Mazur)

There is an anti-equivalence of categories

$$\{p\text{-divisible groups over } k\} \leftrightarrow \left\{ \begin{array}{c} \text{finitely generated free } W(k)\text{-module} \\ M \text{ together with linear maps} \\ F : \sigma^* M \to M,\ V : M \to \sigma^* M \text{ such} \\ \text{that } FV = VF = p \end{array} \right\}$$

Here $\sigma : W(k) \to W(k)$ is the lift of Frobenius.

Dieudonné theory

Dieudonné theory gives a classification of p-divisible groups and finite flat group schemes over a perfect field k of characteristic p in terms of semi-linear algebraic data.

Classical Dieudonné theory (Dieudonné, Grothendieck, Messing, Mazur)

There is an anti-equivalence of categories

$$\{p\text{-divisible groups over } k\} \leftrightarrow \left\{ \begin{array}{c} \text{finitely generated free } W(k)\text{-module} \\ M \text{ together with linear maps} \\ F : \sigma^* M \to M,\ V : M \to \sigma^* M \text{ such} \\ \text{that } FV = VF = p \end{array} \right\}$$

Here $\sigma : W(k) \to W(k)$ is the lift of Frobenius.The objects on the right side are called **Dieudonné modules**. A similar equivalence holds between finite flat group schemes over k and finite $W(k)$-length Dieudonné modules.

Dieudonné theory, cont'd

Examples:

Dieudonné theory, cont'd

Examples:

- The étale group scheme $\mathbb{Z}/p\mathbb{Z}$ has $M(\mathbb{Z}/p\mathbb{Z}) = k$, $F = \sigma$, $V = 0$.

Dieudonné theory, cont'd

Examples:

- The étale group scheme $\mathbb{Z}/p\mathbb{Z}$ has $M(\mathbb{Z}/p\mathbb{Z}) = k$, $F = \sigma$, $V = 0$.
- The multiplicative group scheme μ_p has $M(\mu_p) = k$, $F = 0$, $V = \sigma^{-1}$.

Dieudonné theory, cont'd

Examples:

- The étale group scheme $\mathbb{Z}/p\mathbb{Z}$ has $M(\mathbb{Z}/p\mathbb{Z}) = k$, $F = \sigma$, $V = 0$.
- The multiplicative group scheme μ_p has $M(\mu_p) = k$, $F = 0$, $V = \sigma^{-1}$.
- The additive group scheme α_p has $M(\alpha_p) = k$, $F = V = 0$.

Dieudonné theory, cont'd

Examples:

- The étale group scheme $\mathbb{Z}/p\mathbb{Z}$ has $M(\mathbb{Z}/p\mathbb{Z}) = k$, $F = \sigma$, $V = 0$.
- The multiplicative group scheme μ_p has $M(\mu_p) = k$, $F = 0$, $V = \sigma^{-1}$.
- The additive group scheme α_p has $M(\alpha_p) = k$, $F = V = 0$.
- Finite flat group schemes G over k of rank p^r with $V = 0$ corresponds to the linear map $F : \sigma^* M \to M$ with an r-dimensional k-vector space M.

Dieudonné theory, cont'd

Examples:

- The étale group scheme $\mathbb{Z}/p\mathbb{Z}$ has $M(\mathbb{Z}/p\mathbb{Z}) = k$, $F = \sigma$, $V = 0$.
- The multiplicative group scheme μ_p has $M(\mu_p) = k$, $F = 0$, $V = \sigma^{-1}$.
- The additive group scheme α_p has $M(\alpha_p) = k$, $F = V = 0$.
- Finite flat group schemes G over k of rank p^r with $V = 0$ corresponds to the linear map $F : \sigma^* M \to M$ with an r-dimensional k-vector space M. After fixing a basis, suppose $F = \begin{pmatrix} a_{11} & \cdots & a_{1r} \\ \cdots & \cdots & \cdots \\ a_{r1} & \cdots & a_{rr} \end{pmatrix}$.

Dieudonné theory, cont'd

Examples:

- The étale group scheme $\mathbb{Z}/p\mathbb{Z}$ has $M(\mathbb{Z}/p\mathbb{Z}) = k$, $F = \sigma$, $V = 0$.
- The multiplicative group scheme μ_p has $M(\mu_p) = k$, $F = 0$, $V = \sigma^{-1}$.
- The additive group scheme α_p has $M(\alpha_p) = k$, $F = V = 0$.
- Finite flat group schemes G over k of rank p^r with $V = 0$ corresponds to the linear map $F : \sigma^* M \to M$ with an r-dimensional k-vector space M. After fixing a basis, suppose $F = \begin{pmatrix} a_{11} & \cdots & a_{1r} \\ \cdots & \cdots & \cdots \\ a_{r1} & \cdots & a_{rr} \end{pmatrix}$. Then $G = \operatorname{Spec} k[x_1, \cdots, x_r]/(\{x_i^p - \sum_{j=1}^r a_{ji} x_j\}_{1 \le i \le r})$ with the Hopf algebra structure $m(x_i) = 1 \otimes x_i + x_i \otimes 1$, $\operatorname{inv}(x_i) = -x_i$, $e(x_i) = 0$.

Dieudonné theory over a mixed-characteristic base

More generally, one constructs a natural functor from left to right, **Dieudonné functor** in various cases (Berthelot, Breen, Messing, de Jong, Fontaine, Breuil, Kisin, Kim, Cais, Lau, Fargues, Scholze, Weinstein, $\cdots$).

Dieudonné theory over a mixed-characteristic base

More generally, one constructs a natural functor from left to right, **Dieudonné functor** in various cases (Berthelot, Breen, Messing, de Jong, Fontaine, Breuil, Kisin, Kim, Cais, Lau, Fargues, Scholze, Weinstein, $\cdots$). If the base field k is replaced by a ring R, then often there is another ring A that plays the role of $W(k)$. One has to have R as a quotient of A, just like $W(k)/p = k$.

Dieudonné theory over a mixed-characteristic base

More generally, one constructs a natural functor from left to right, **Dieudonné functor** in various cases (Berthelot, Breen, Messing, de Jong, Fontaine, Breuil, Kisin, Kim, Cais, Lau, Fargues, Scholze, Weinstein, $\cdots$). If the base field k is replaced by a ring R, then often there is another ring A that plays the role of $W(k)$. One has to have R as a quotient of A, just like $W(k)/p = k$.

Example (Breuil–Kisin modules)

Dieudonné theory over a mixed-characteristic base

More generally, one constructs a natural functor from left to right, **Dieudonné functor** in various cases (Berthelot, Breen, Messing, de Jong, Fontaine, Breuil, Kisin, Kim, Cais, Lau, Fargues, Scholze, Weinstein, $\cdots$). If the base field k is replaced by a ring R, then often there is another ring A that plays the role of $W(k)$. One has to have R as a quotient of A, just like $W(k)/p = k$.

Example (Breuil–Kisin modules)

Let $R = \mathcal{O}_K$ for $K/\mathbb{Q}_p$ a finite extension, with residue field k and uniformizer π.

Dieudonné theory over a mixed-characteristic base

More generally, one constructs a natural functor from left to right, **Dieudonné functor** in various cases (Berthelot, Breen, Messing, de Jong, Fontaine, Breuil, Kisin, Kim, Cais, Lau, Fargues, Scholze, Weinstein, $\cdots$). If the base field k is replaced by a ring R, then often there is another ring A that plays the role of $W(k)$. One has to have R as a quotient of A, just like $W(k)/p = k$.

Example (Breuil–Kisin modules)

Let $R = \mathcal{O}_K$ for $K/\mathbb{Q}_p$ a finite extension, with residue field k and uniformizer π. There is a classification of p-divisible groups over $\mathcal{O}_K$, with the role of $W(k)$ played by $\mathfrak{S} := W(k)[[u]]$, with the quotient map $\mathfrak{S} \overset{u \mapsto \pi}{\twoheadrightarrow} \mathcal{O}_K$.

Dieudonné theory over a mixed-characteristic base

More generally, one constructs a natural functor from left to right, **Dieudonné functor** in various cases (Berthelot, Breen, Messing, de Jong, Fontaine, Breuil, Kisin, Kim, Cais, Lau, Fargues, Scholze, Weinstein, $\cdots$). If the base field k is replaced by a ring R, then often there is another ring A that plays the role of $W(k)$. One has to have R as a quotient of A, just like $W(k)/p = k$.

Example (Breuil–Kisin modules)

Let $R = \mathcal{O}_K$ for $K/\mathbb{Q}_p$ a finite extension, with residue field k and uniformizer π. There is a classification of p-divisible groups over $\mathcal{O}_K$, with the role of $W(k)$ played by $\mathfrak{S} := W(k)[[u]]$, with the quotient map $\mathfrak{S} \overset{u \mapsto \pi}{\twoheadrightarrow} \mathcal{O}_K$. There is an anti-equivalence of categories

Dieudonné theory over a mixed-characteristic base

More generally, one constructs a natural functor from left to right, **Dieudonné functor** in various cases (Berthelot, Breen, Messing, de Jong, Fontaine, Breuil, Kisin, Kim, Cais, Lau, Fargues, Scholze, Weinstein, $\cdots$). If the base field k is replaced by a ring R, then often there is another ring A that plays the role of $W(k)$. One has to have R as a quotient of A, just like $W(k)/p = k$.

Example (Breuil–Kisin modules)

Let $R = \mathcal{O}_K$ for $K/\mathbb{Q}_p$ a finite extension, with residue field k and uniformizer π. There is a classification of p-divisible groups over $\mathcal{O}_K$, with the role of $W(k)$ played by $\mathfrak{S} := W(k)[[u]]$, with the quotient map $\mathfrak{S} \overset{u \mapsto \pi}{\twoheadrightarrow} \mathcal{O}_K$. There is an anti-equivalence of categories

$$\{p\text{-divisible groups over } \mathcal{O}_K\} \leftrightarrow \left\{ \begin{array}{c} \text{Finite free } \mathfrak{S}\text{-modules } \mathfrak{M} \text{ with a linear} \\ \text{map } F : \varphi^*\mathfrak{M} \to \mathfrak{M} \text{ whose cokernel is} \\ \text{killed by } E(u) \end{array} \right\}$$

Dieudonné theory over a mixed-characteristic base

More generally, one constructs a natural functor from left to right, **Dieudonné functor** in various cases (Berthelot, Breen, Messing, de Jong, Fontaine, Breuil, Kisin, Kim, Cais, Lau, Fargues, Scholze, Weinstein, $\cdots$). If the base field k is replaced by a ring R, then often there is another ring A that plays the role of $W(k)$. One has to have R as a quotient of A, just like $W(k)/p = k$.

Example (Breuil–Kisin modules)

Let $R = \mathcal{O}_K$ for $K/\mathbb{Q}_p$ a finite extension, with residue field k and uniformizer π. There is a classification of p-divisible groups over $\mathcal{O}_K$, with the role of $W(k)$ played by $\mathfrak{S} := W(k)[[u]]$, with the quotient map $\mathfrak{S} \overset{u \mapsto \pi}{\twoheadrightarrow} \mathcal{O}_K$. There is an anti-equivalence of categories

$$\{p\text{-divisible groups over } \mathcal{O}_K\} \leftrightarrow \left\{ \begin{array}{c} \text{Finite free } \mathfrak{S}\text{-modules } \mathfrak{M} \text{ with a linear} \\ \text{map } F : \varphi^*\mathfrak{M} \to \mathfrak{M} \text{ whose cokernel is} \\ \text{killed by } E(u) \end{array} \right\}$$

Here $E(u)$ is the minimal Eisenstein polynomial of π over $W(k)$ (a generator of $\ker(\mathfrak{S} \to \mathcal{O}_K)$), and φ extends σ with $\varphi(u) = u^p$.

Prismatic Dieudonné theory

These two constructions use some pair (A, I) such that A/I is the base ring R and A is equipped with a lift of Frobenius $\varphi : A \to A$.

Prismatic Dieudonné theory

These two constructions use some pair (A, I) such that A/I is the base ring R and A is equipped with a lift of Frobenius $\varphi : A \to A$. The pairs $(W(k), (p))$ and $(\mathfrak{S}, (E(u)))$ are examples of **prisms**, and the notion of prisms is considered as the essence of all p-adic cohomology theories.

Prismatic Dieudonné theory

These two constructions use some pair (A, I) such that A/I is the base ring R and A is equipped with a lift of Frobenius $\varphi : A \to A$. The pairs $(W(k), (p))$ and $(\mathfrak{S}, (E(u)))$ are examples of **prisms**, and the notion of prisms is considered as the essence of all p-adic cohomology theories.

Theorem (Anschütz–Le Bras)

Let R be a p-complete Noetherian local complete intersection ring.

Prismatic Dieudonné theory

These two constructions use some pair (A, I) such that A/I is the base ring R and A is equipped with a lift of Frobenius $\varphi : A \to A$. The pairs $(W(k), (p))$ and $(\mathfrak{S}, (E(u)))$ are examples of **prisms**, and the notion of prisms is considered as the essence of all p-adic cohomology theories.

Theorem (Anschütz–Le Bras)

Let R be a p-complete Noetherian local complete intersection ring. Then, there is an anti-equivalence between

$$\{p\text{-divisible groups over } R\} \leftrightarrow \left\{ \begin{array}{c} \text{Finite free } A\text{-modules } \mathcal{M} \text{ with a linear} \\ \text{map } \varphi_{\mathcal{M}} : \varphi^* \mathcal{M} \to \mathcal{M} \text{ whose} \\ \text{cokernel is killed by } I \text{ satisfying the} \\ \text{admissibility condition} \end{array} \right\}$$

Prismatic Dieudonné theory

These two constructions use some pair (A, I) such that A/I is the base ring R and A is equipped with a lift of Frobenius $\varphi : A \to A$. The pairs $(W(k), (p))$ and $(\mathfrak{S}, (E(u)))$ are examples of **prisms**, and the notion of prisms is considered as the essence of all p-adic cohomology theories.

Theorem (Anschütz–Le Bras)

Let R be a p-complete Noetherian local complete intersection ring. Then, there is an anti-equivalence between

$$\{p\text{-divisible groups over } R\} \leftrightarrow \left\{ \begin{array}{c} \text{Finite free } A\text{-modules } \mathcal{M} \text{ with a linear} \\ \text{map } \varphi_{\mathcal{M}} : \varphi^* \mathcal{M} \to \mathcal{M} \text{ whose} \\ \text{cokernel is killed by } I \text{ satisfying the} \\ \text{admissibility condition} \end{array} \right\}$$

The objects in the right side are called **admissible prismatic Dieudonné crystals**.

Prismatic Dieudonné theory

These two constructions use some pair (A, I) such that A/I is the base ring R and A is equipped with a lift of Frobenius $\varphi : A \to A$. The pairs $(W(k), (p))$ and $(\mathfrak{S}, (E(u)))$ are examples of **prisms**, and the notion of prisms is considered as the essence of all p-adic cohomology theories.

Theorem (Anschütz–Le Bras)

Let R be a p-complete Noetherian local complete intersection ring. Then, there is an anti-equivalence between

$$\{p\text{-divisible groups over } R\} \leftrightarrow \left\{ \begin{array}{c} \text{Finite free } A\text{-modules } \mathcal{M} \text{ with a linear} \\ \text{map } \varphi_{\mathcal{M}} : \varphi^* \mathcal{M} \to \mathcal{M} \text{ whose} \\ \text{cokernel is killed by } I \text{ satisfying the} \\ \text{admissibility condition} \end{array} \right\}$$

The objects in the right side are called **admissible prismatic Dieudonné crystals**.

In reality, prismatic crystals are sheaves over the absolute prismatic site $(R)_{\Delta}$, so they should be thought as the limit of such objects over varying (A, I).

Prismatic Dieudonné theory

These two constructions use some pair (A, I) such that A/I is the base ring R and A is equipped with a lift of Frobenius $\varphi : A \to A$. The pairs $(W(k), (p))$ and $(\mathfrak{S}, (E(u)))$ are examples of **prisms**, and the notion of prisms is considered as the essence of all p-adic cohomology theories.

Theorem (Anschütz–Le Bras)

Let R be a p-complete Noetherian local complete intersection ring. Then, there is an anti-equivalence between

$$\{p\text{-divisible groups over } R\} \leftrightarrow \left\{ \begin{array}{c} \text{Finite free } A\text{-modules } \mathcal{M} \text{ with a linear} \\ \text{map } \varphi_{\mathcal{M}} : \varphi^* \mathcal{M} \to \mathcal{M} \text{ whose} \\ \text{cokernel is killed by } I \text{ satisfying the} \\ \text{admissibility condition} \end{array} \right\}$$

The objects in the right side are called **admissible prismatic Dieudonné crystals**.

In reality, prismatic crystals are sheaves over the absolute prismatic site $(R)_{\mathbb{\Delta}}$, so they should be thought as the limit of such objects over varying (A, I). If R is perfectoid, there is a final object $(A_{\mathrm{inf}}(R), \ker(\widetilde{\theta}))$ in $(R)_{\mathbb{\Delta}}$.

Prismatic Dieudonné theory

These two constructions use some pair (A, I) such that A/I is the base ring R and A is equipped with a lift of Frobenius $\varphi : A \to A$. The pairs $(W(k), (p))$ and $(\mathfrak{S}, (E(u)))$ are examples of **prisms**, and the notion of prisms is considered as the essence of all p-adic cohomology theories.

Theorem (Anschütz–Le Bras)

Let R be a p-complete Noetherian local complete intersection ring. Then, there is an anti-equivalence between

$$\{p\text{-divisible groups over } R\} \leftrightarrow \left\{ \begin{array}{c} \text{Finite free } A\text{-modules } \mathcal{M} \text{ with a linear} \\ \text{map } \varphi_{\mathcal{M}} : \varphi^* \mathcal{M} \to \mathcal{M} \text{ whose} \\ \text{cokernel is killed by } I \text{ satisfying the} \\ \text{admissibility condition} \end{array} \right\}$$

The objects in the right side are called **admissible prismatic Dieudonné crystals**.

In reality, prismatic crystals are sheaves over the absolute prismatic site $(R)_{\mathbb{\Delta}}$, so they should be thought as the limit of such objects over varying (A, I). If R is perfectoid, there is a final object $(A_{\mathrm{inf}}(R), \ker(\widetilde{\theta}))$ in $(R)_{\mathbb{\Delta}}$. In this talk, we will pretend that there is such final object (A, I), and that there is a Frobenius on R.

Outline

1. Integral models of Shimura varieties
2. (Prismatic) Dieudonné theory
3. Level structures via (prismatic) Dieudonné theory

$\Gamma_0(p)$, $\Gamma_1(p)$, $\Gamma(p)$

We now suggest the notion of level structures using (prismatic) Dieudonné theory.

$\Gamma_0(p)$, $\Gamma_1(p)$, $\Gamma(p)$

We now suggest the notion of level structures using (prismatic) Dieudonné theory.

Definition

For a polarized Dieudonné module $\mathcal{M}$ over A, free of rank $2g$, the $\Gamma_0(p)$-level is

$$L: \quad \mathcal{M} \twoheadrightarrow \mathcal{H}_g \twoheadrightarrow \mathcal{H}_{g-1} \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{H}_1 \twoheadrightarrow \mathcal{H}_0 = 0,$$

$\Gamma_0(p)$, $\Gamma_1(p)$, $\Gamma(p)$

We now suggest the notion of level structures using (prismatic) Dieudonné theory.

Definition

For a polarized Dieudonné module $\mathcal{M}$ over A, free of rank $2g$, the $\Gamma_0(p)$-level is

$$L: \quad \mathcal{M} \twoheadrightarrow \mathcal{H}_g \twoheadrightarrow \mathcal{H}_{g-1} \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{H}_1 \twoheadrightarrow \mathcal{H}_0 = 0,$$

where $\mathcal{D}_i := \ker(\mathcal{H}_i \twoheadrightarrow \mathcal{H}_{i-1})$ is a free R-module of rank 1, such that the compatibility with polarization is satisfied.

$\Gamma_0(p)$, $\Gamma_1(p)$, $\Gamma(p)$

We now suggest the notion of level structures using (prismatic) Dieudonné theory.

Definition

For a polarized Dieudonné module $\mathcal{M}$ over A, free of rank $2g$, the $\Gamma_0(p)$-level is

$$L: \quad \mathcal{M} \twoheadrightarrow \mathcal{H}_g \twoheadrightarrow \mathcal{H}_{g-1} \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{H}_1 \twoheadrightarrow \mathcal{H}_0 = 0,$$

where $\mathcal{D}_i := \ker(\mathcal{H}_i \twoheadrightarrow \mathcal{H}_{i-1})$ is a free R-module of rank 1, such that the compatibility with polarization is satisfied.

The $\Gamma_1(p)$-level is such that there are maps of torsion Dieudonné modules $f_i : (R, \sigma) \to \mathcal{D}_i$ such that $\operatorname{coker} f_i \cong \operatorname{coker}(\varphi : \varphi^* \mathcal{D}_i \to \mathcal{D}_i)$.

$\Gamma_0(p)$, $\Gamma_1(p)$, $\Gamma(p)$

We now suggest the notion of level structures using (prismatic) Dieudonné theory.

Definition

For a polarized Dieudonné module $\mathcal{M}$ over A, free of rank $2g$, the $\Gamma_0(p)$-level is

$$L: \quad \mathcal{M} \twoheadrightarrow \mathcal{H}_g \twoheadrightarrow \mathcal{H}_{g-1} \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{H}_1 \twoheadrightarrow \mathcal{H}_0 = 0,$$

where $\mathcal{D}_i := \ker(\mathcal{H}_i \twoheadrightarrow \mathcal{H}_{i-1})$ is a free R-module of rank 1, such that the compatibility with polarization is satisfied.

The $\Gamma_1(p)$-level is such that there are maps of torsion Dieudonné modules $f_i : (R, \sigma) \to \mathcal{D}_i$ such that $\operatorname{coker} f_i \cong \operatorname{coker}(\varphi : \varphi^* \mathcal{D}_i \to \mathcal{D}_i)$.

The $\Gamma(p)$-level is such that there is a map of chains from the $\Gamma_0(p)$-level of the multiplicative Dieudonné module $A^{\oplus 2g} \twoheadrightarrow \mathcal{C}_g \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{C}_1$ to L such that the induced maps $\ker(\mathcal{C}_i \twoheadrightarrow \mathcal{C}_{i-1}) \to \mathcal{D}_i$ give rise to a $\Gamma_1(p)$-structure.

$\Gamma_0(p)$, $\Gamma_1(p)$, $\Gamma(p)$

We now suggest the notion of level structures using (prismatic) Dieudonné theory.

Definition

For a polarized Dieudonné module $\mathcal{M}$ over A, free of rank $2g$, the $\Gamma_0(p)$-level is

$$L: \quad \mathcal{M} \twoheadrightarrow \mathcal{H}_g \twoheadrightarrow \mathcal{H}_{g-1} \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{H}_1 \twoheadrightarrow \mathcal{H}_0 = 0,$$

where $\mathcal{D}_i := \ker(\mathcal{H}_i \twoheadrightarrow \mathcal{H}_{i-1})$ is a free R-module of rank 1, such that the compatibility with polarization is satisfied.

The $\Gamma_1(p)$-level is such that there are maps of torsion Dieudonné modules $f_i : (R, \sigma) \to \mathcal{D}_i$ such that $\operatorname{coker} f_i \cong \operatorname{coker}(\varphi : \varphi^* \mathcal{D}_i \to \mathcal{D}_i)$.

The $\Gamma(p)$-level is such that there is a map of chains from the $\Gamma_0(p)$-level of the multiplicative Dieudonné module $A^{\oplus 2g} \twoheadrightarrow \mathcal{C}_g \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{C}_1$ to L such that the induced maps $\ker(\mathcal{C}_i \twoheadrightarrow \mathcal{C}_{i-1}) \to \mathcal{D}_i$ give rise to a $\Gamma_1(p)$-structure.

Theorem (O.)

The moduli of the above level structures is flat over the base.

Deeper levels

We tentatively suggest the following definition, following the work of Bieker.

Definition

For a polarized p-divisible group $\mathcal{G}$ over R of height $2g$, the **naive $\Gamma_0(p^n)$-level** is

$$L: \quad \mathcal{M}(\mathcal{G}) \twoheadrightarrow \mathcal{H}_g \twoheadrightarrow \mathcal{H}_{g-1} \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{H}_1 \twoheadrightarrow \mathcal{H}_0 = 0,$$

Deeper levels

We tentatively suggest the following definition, following the work of Bieker.

Definition

For a polarized p-divisible group $\mathcal{G}$ over R of height $2g$, the **naive $\Gamma_0(p^n)$-level** is

$$L: \quad \mathcal{M}(\mathcal{G}) \twoheadrightarrow \mathcal{H}_g \twoheadrightarrow \mathcal{H}_{g-1} \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{H}_1 \twoheadrightarrow \mathcal{H}_0 = 0,$$

where $\mathcal{D}_i := \ker(\mathcal{H}_i \twoheadrightarrow \mathcal{H}_{i-1})$ is a free A/I^n-module of rank 1, such that the compatibility with polarization is satisfied.

Deeper levels

We tentatively suggest the following definition, following the work of Bieker.

Definition

For a polarized p-divisible group $\mathcal{G}$ over R of height $2g$, the **naive $\Gamma_0(p^n)$-level** is

$$L: \quad \mathcal{M}(\mathcal{G}) \twoheadrightarrow \mathcal{H}_g \twoheadrightarrow \mathcal{H}_{g-1} \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{H}_1 \twoheadrightarrow \mathcal{H}_0 = 0,$$

where $\mathcal{D}_i := \ker(\mathcal{H}_i \twoheadrightarrow \mathcal{H}_{i-1})$ is a free A/I^n-module of rank 1, such that the compatibility with polarization is satisfied.

The integral model of abelian varieties with $\Gamma_0(p^n)$-level structures $\mathfrak{A}_{g,\Gamma_0(p^n)}$ over R is the scheme-theoretic closure of

$$\mathfrak{A}_{g,\Gamma_0(p^n),\text{naive}} \hookrightarrow \varprojlim_{P \supset \Gamma_0(p^n)} \mathfrak{A}_{g,P},$$

where the limit is taken over all parahoric subgroups containing $\Gamma_0(p^n)$.

Deeper levels

We tentatively suggest the following definition, following the work of Bieker.

Definition

For a polarized p-divisible group $\mathcal{G}$ over R of height $2g$, the **naive $\Gamma_0(p^n)$-level** is

$$L: \quad \mathcal{M}(\mathcal{G}) \twoheadrightarrow \mathcal{H}_g \twoheadrightarrow \mathcal{H}_{g-1} \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{H}_1 \twoheadrightarrow \mathcal{H}_0 = 0,$$

where $\mathcal{D}_i := \ker(\mathcal{H}_i \twoheadrightarrow \mathcal{H}_{i-1})$ is a free A/I^n-module of rank 1, such that the compatibility with polarization is satisfied.

The integral model of abelian varieties with $\Gamma_0(p^n)$-level structures $\mathfrak{A}_{g,\Gamma_0(p^n)}$ over R is the scheme-theoretic closure of

$$\mathfrak{A}_{g,\Gamma_0(p^n),\text{naive}} \hookrightarrow \varprojlim_{P \supset \Gamma_0(p^n)} \mathfrak{A}_{g,P},$$

where the limit is taken over all parahoric subgroups containing $\Gamma_0(p^n)$.

For $R/\mathbb{Z}_p$, this is the flat closure of the generic fiber, and the formation of moduli scheme commutes with flat base change.

Example: $\Gamma_0(p^2)$, $\Gamma_1(p^2)$

For an elliptic curve E over $\mathcal{O}_{\mathbb{C}_p}$, the naive $\Gamma_0(p^2)$-level of $E[p^\infty]$ is

$$\mathcal{M}_2(E[p^\infty]) \twoheadrightarrow \mathcal{H},$$

where $\mathcal{H}$ is a free $W_2(\mathcal{O}^\flat_{\mathbb{C}_p})$-module of rank 1.

Example: $\Gamma_0(p^2)$, $\Gamma_1(p^2)$

For an elliptic curve E over $\mathcal{O}_{\mathbb{C}_p}$, the naive $\Gamma_0(p^2)$-level of $E[p^\infty]$ is

$$\mathcal{M}_2(E[p^\infty]) \twoheadrightarrow \mathcal{H},$$

where $\mathcal{H}$ is a free $W_2(\mathcal{O}^\flat_{\mathbb{C}_p})$-module of rank 1. You can then get two different $\Gamma_0(p)$-levels,

$$\mathcal{M}_1(E[p^\infty]) \twoheadrightarrow \mathcal{H}/p, \quad \mathcal{M}_1(E[p^\infty]/H) \twoheadrightarrow \varphi^*\mathcal{M}_1(E[p^\infty]).$$

Example: $\Gamma_0(p^2)$, $\Gamma_1(p^2)$

For an elliptic curve E over $\mathcal{O}_{\mathbb{C}_p}$, the naive $\Gamma_0(p^2)$-level of $E[p^\infty]$ is

$$\mathcal{M}_2(E[p^\infty]) \twoheadrightarrow \mathcal{H},$$

where $\mathcal{H}$ is a free $W_2(\mathcal{O}^\flat_{\mathbb{C}_p})$-module of rank 1. You can then get two different $\Gamma_0(p)$-levels,

$$\mathcal{M}_1(E[p^\infty]) \twoheadrightarrow \mathcal{H}/p, \quad \mathcal{M}_1(E[p^\infty]/H) \twoheadrightarrow \varphi^*\mathcal{M}_1(E[p^\infty]).$$

The $\Gamma_1(p^2)$-level is an additional data of torsion Dieudonné modules over $R[\mathcal{H}]$, $f : (R[\mathcal{H}], \sigma) \to \mathcal{H}[p]_{R[\mathcal{H}]}$, such that $\operatorname{coker} f_i \cong \operatorname{coker} \varphi^*$ (Dieudonné version of primitive elements by Kottwitz–Wake).

Thank you for listening!